# บทที่ 3 การออกแบบคลาส

# (Class Design)

**รายละเอียดของคลาส**

| MainClass |
| --- |
| + void main (String [] args) |

| LoadGame |
| --- |
| - int SCREEN_WIDTH<br>- int SCREEN_HEIGHT<br>- int currentIndex<br>- float animationPos<br>- BufferedImage loadBg<br>- ImageIcon[] animationG<br>- ImageIcon[] runAnimation<br>- AudioInputStream loadGameMusic<br>- Clip loadGameC<br>- Timer lRunAnimations<br>- Timer waitL<br>- Timer swapCha |
| + LoadGame ()<br>- void drawProgressBar (Graphics g)<br>- void loadGameBG ()<br>+ void loadBeforeToLobby ()<br>+ void actionPerformed (ActionEvent e) |

| LobbyUI | |
|---|---|
| - int SCREEN_WIDTH<br>- int SCREEN_HEIGHT<br>+ boolean IsVolUpClicked<br>+ boolean IsVolUp2Clicked<br>+ boolean IsVBClicked<br>+ boolean IsVB2Clicked<br>+ boolean IsDragging<br>+ boolean IsBoughtCat<br>+ boolean IsBoughtChic<br>+ boolean IsBoughtSlime<br>- JFrame jframe<br>- JPanel panel, panel2<br>+ JPanel panel3<br>- JLabel label<br>+ JLabel scoreboard, balances<br>- JButton button, button2, button3, yesButton, noButton, volUpButton, volUpButton2, backButton<br>+ JButton volPButton, volPButton2, dogButton, catButton, chickenButton, slimeButton<br>+ BufferedImage img, img2, img3, yImg, nImage, volImage, volZImage, volPImage, lobbybg,flyingbg, quitbg, settingbg, backImage,dog<br>+ BufferedImage [] slime, chicken, cat<br>+ ImageIcon[] showingCha<br>- ImageIcon showCharacter | + ImageIcon dogIcon, dogIcon2, slimeIcon, slimeIcon2, chicIcon, chicIcon2, catIcon, catIcon2<br>+ ImageIcon playButtonIcon, quitButton, settingButton, yesButtonIcon, noButtonIcon, volImageIcon, volPIcon, bImageIcon<br>+ Image resizedImg, resizedImg2, resizedImg3, resizedImg4, resizedImg5, resizedImage7, resizedImage8<br>+ Image[] resizedImage6<br>- AudioInputStream LobbyMusic, clickSound<br>- Clip lobbyC, clickC<br>+ float targetVolume, targetVolume2<br>+ int CharacterIndex, totalBalance, max_scored<br>- float animationPos<br>- Timer movingBG<br><br>+ LobbyUI()<br>- void loadCharacter()<br>+ void loadLobbyMusic()<br>- void clickPlay()<br>+ void setVolPButtonX(int x)<br>+ void setVolPButtonX2(int x)<br>+ void actionPerformed(ActionEvent e) |

| MouseInputs |
|---|
| - LobbyUI lobbyUI<br>- int initial<br>- int initialX2 |
| + MouseInputs (LobbyUI lobbyUI)<br>+ void mouseClicked (MouseEvent e)<br>+ void mousePressed (MouseEvent e)<br>+ void mouseReleased (MouseEvent e)<br>+ void mouseEntered (MouseEvent e)<br>+ void mouseExited (MouseEvent e)<br>+ void mouseDragged (MouseEvent e)<br>+ void mouseMoved(MouseEvent e) |

| KeyboardInputs |
| --- |
| - GamePanel gamePanel<br>- Player player<br>- int CharacterIndex<br>- float targetVolume2<br>- Clip CharacterJC, CharacterSC<br>- boolean pressed<br>+ AudioInputStream CharacterJS<br>+ AudioInputStream CharacterSS |
| + KeyboardInputs(GamePanel gamePanel, int CharacterIndex, float targetVolume2)<br>- void playJSound ()<br>- void playSSound ()<br>+ void keyTyped (KeyEvent e)<br>+ void keyPressed (KeyEvent e)<br>+ void keyReleased (KeyEvent e) |

| LoadBeforeRun |
| --- |
| - int SCREEN_WIDTH<br>- int SCREEN_HEIGHT<br>- int currentIndex<br>- BufferedImage loadBeforeRunBG<br>- BufferedImage characterRun<br>- AudioInputStream loadGameMusic<br>- Clip loadGameC<br>- int CharacterIndex<br>- float targetVolume2<br>- String text = "" |
| + LoadBeforeRun(int CharacterIndex, float targetVolume2, float targetVolume)<br>- void loadBeforeRunBG ()<br>+ void loadBeforeToRun ()<br>+ void actionPerformed(ActionEvent e) |

| LevelManager |
| --- |
| - Game game<br>+ int RandomLvl<br>+ Random rand<br>- MoneyPoint[] points<br>- Obstacle[] obstacles<br>- Item[] item |
| + Level Manager (Game game)<br>+ void randomMap ()<br>- void lvl1 ()<br>- void lvl2 ()<br>- void lvl3 ()<br>- void lvl4 ()<br>- void lvl5 ()<br>+ void lvl5 ()<br>+ void update ()<br>+ void render (Graphics g) |

| GameWindow |
| --- |
| + JFrame jframe |
| + GameWindow (GamePanel gamePanel) |

| LoadAssets |
| --- |
| - GamePanel gamePanel<br>- Player player<br>- LobbyUI lobbyUI<br>- BufferedImage img<br>- AudioInputStream audioStream<br>- String petName |
| + LoadAssets(GamePanel gamePanel)<br>+ LoadAssets(Player player)<br>+ LoadAssets(Item item)<br>+ LoadAssets(Obstacle obstacle)<br>+ LoadAssets(MoneyPoint moneyPoint)<br>+ LoadAssets(LobbyUI lobbyUI)<br>- BufferedImage readImageIO(String pathString)<br>- AudioInputStream readAudioFile(String pathString)<br>- void loadLobbyPics()<br>- void loadAnimations()<br>- void loadBackground()<br>- void loadSFXs() |

## GamePanel

```
- Game game
+ int SCREEN_WIDTH
+ int SCREEN_HEIGHT
+ double BG_X
+ BufferedImage forestBackground
+ BufferedImage sandBackground
+ BufferedImage snowBackground
+ BufferedImage duck1, coin1
- boolean isBG1
- boolean isBG2
+ boolean forestMap
+ boolean sandMap
+ boolean snowMap
+ boolean Isplay
+ BufferedImage currBackground1
+ BufferedImage currBackground2
+ BufferedImage HPbar
+ AudioInputStream desertTheme
+ AudioInputStream snowTheme
+ AudioInputStream forestTheme
+ Clip clip
- float targetVolume
+ double HP
+ boolean IsInvi
+ int totalScore
+ int totalCoin
+ JLabel showScore, showCoin
```

```
+ GamePanel(Game game, int indexType, float
  targetVolume2, float targetVolume)
- void setPanelSize()
+ void updateGamePanel()
+ void paintComponent(Graphics g)
+ void playSound()
+ Game getGame()
+ void loadAndReadFile ()
```

| Game (implements Runnable) |
| --- |
| - GameWindow gameWindow<br>+ GamePanel gamePanel<br>- Thread gameThread<br>+ LevelManager levelManager<br>- Player player<br><br>+ final int SCREEN_WIDTH<br>+ final int SCREEN_HEIGHT<br>- final int FPS_SET<br>- final int UPS_SET<br><br>- final static int TILE_SIZE<br>- final int PLAYER_X<br>- final int PLAYER_Y<br>- String type;<br><br>- int indexType<br>- float Volume1, Volume2<br>- int max_scored, balances |
| + Game(int CharacterIndex, float targetVolume2, float targetVolume, int max_scored, int balances)<br>- void getStringType()<br>- void startGameLoop()<br>+ void update()<br>+ void render()<br>+ @Override void run()<br>+ Player getPlayer() |

| GameObject |
| --- |
| + int x,y<br>- int width,height<br>+ int offsetX,offsetY,offsetWidth,offsetHeight<br>+ boolean active |
| + GameObject(int x, int y, int width, int height)<br>- void initHitbox()<br>+ void setActiveObject(int x, int y) |

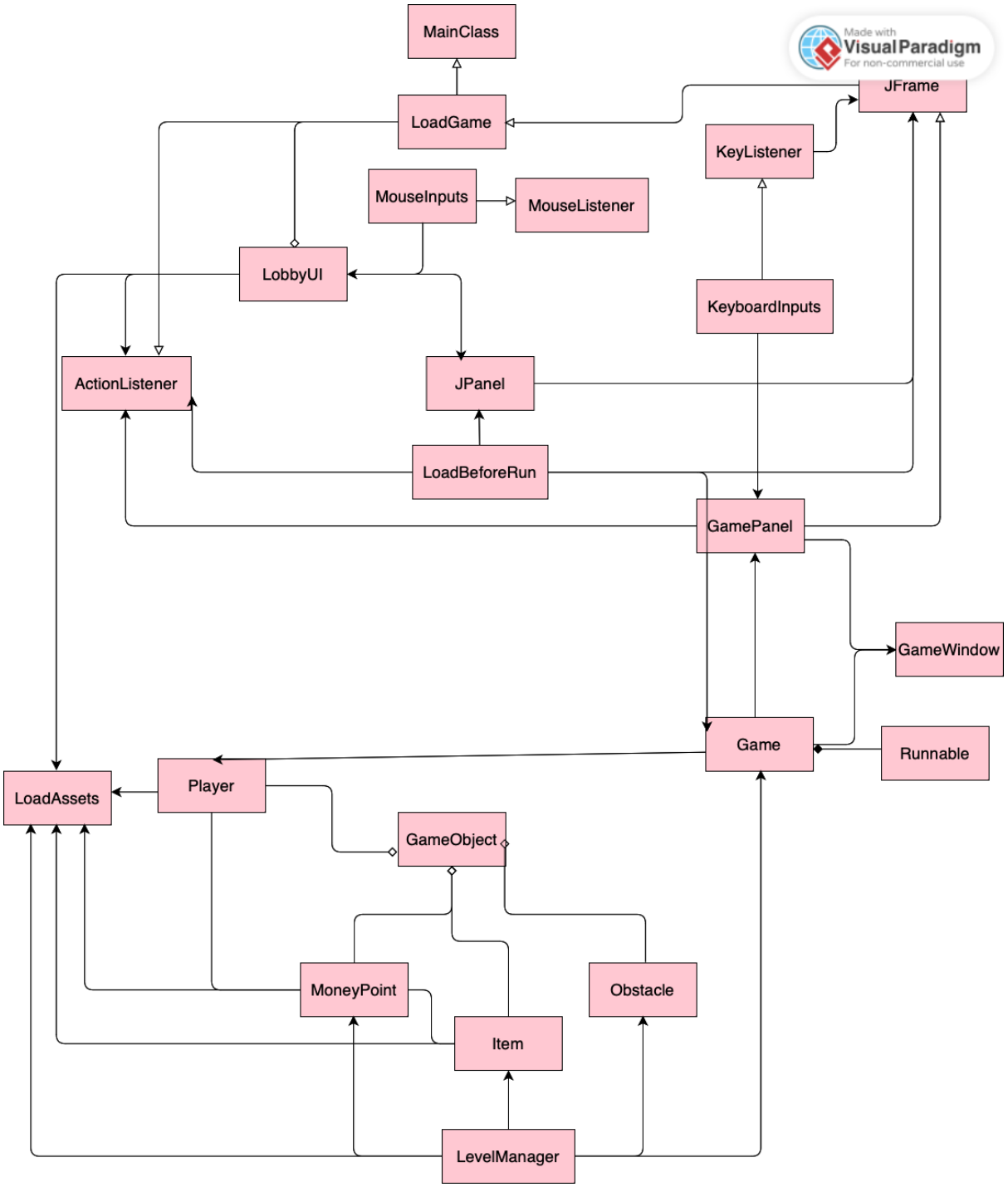
| Item |
| --- |
| - Game game<br>+ BufferedImage img<br>- AudioInputStream potionS<br>- Clip potionC |
| + Item(int x, int y, int width, int height, Game game)<br>+ void update()<br>+ void render(Graphics g)<br>+ void checkHitbox()<br>- void playHitSound() |

| MoneyPoint |
| --- |
| - Game game<br>+ String name<br>+ BufferedImage cake<br>+ BufferedImage duck<br>+ BufferedImage gold<br>+ BufferedImage img<br>- AudioInputStream pointS<br>- Clip pointC |
| + MoneyPoint(String name, int x, int y, int width, int height, Game game)<br>+ void update()<br>+ void render(Graphics g)<br>+ void checkHitbox()<br>- void playHitSound() |

| Obstacle |
| --- |
| - Game game<br>+ BufferedImage forestFloor<br>+ BufferedImage sandFloor<br>+ BufferedImage snowFloor<br>+ BufferedImage rock<br>+ BufferedImage new_sunflower<br>+ BufferedImage new_cactus<br>+ BufferedImage new_sandCastle<br>+ BufferedImage snowman<br>+ BufferedImage krasue<br>+ BufferedImage new_bee<br>+ BufferedImage new_mallard<br>+ BufferedImage ice<br>+ BufferedImage img |
| + Obstacle(String name, int x, int y, int width, int height, Game game)<br>+ void update()<br>+ void render(Graphics g)<br>+ void checkHitbox() |

| Player |
| --- |
| + String type<br>+ BufferedImage img<br>+ int NUM_FRAMES_RUN<br>+ int NUM_FRAMES_JUMP<br>+ int NUM_FRAMES_SLIDE<br>+ BufferedImage[] runningFrames<br>+ BufferedImage[] jumpingFrames<br>+ BufferedImage[] slidingFrames<br>- int currentFrameIndex<br>- int currentFrameIndex3<br>+ int JUMP_HEIGHT<br>+ int yDelta<br>+ int addY<br>+ int jumpCount<br>+ boolean canDoubleJump<br>+ boolean isUp<br>+ boolean inAir<br>+ boolean isJump<br>+ boolean isSlide<br>+ int playerSpeed<br>+ int PLAYER_SPEED_LIMIT<br>+ int tickSpeed<br>+ int MAX_TICK_SPEED |
| + Player(int x, int y, int width, int height, String type)<br>+ void update()<br>+ void render(Graphics g)<br>+ void nextFrame()<br>+ void run()<br>+ void slide()<br>+ void jump() |

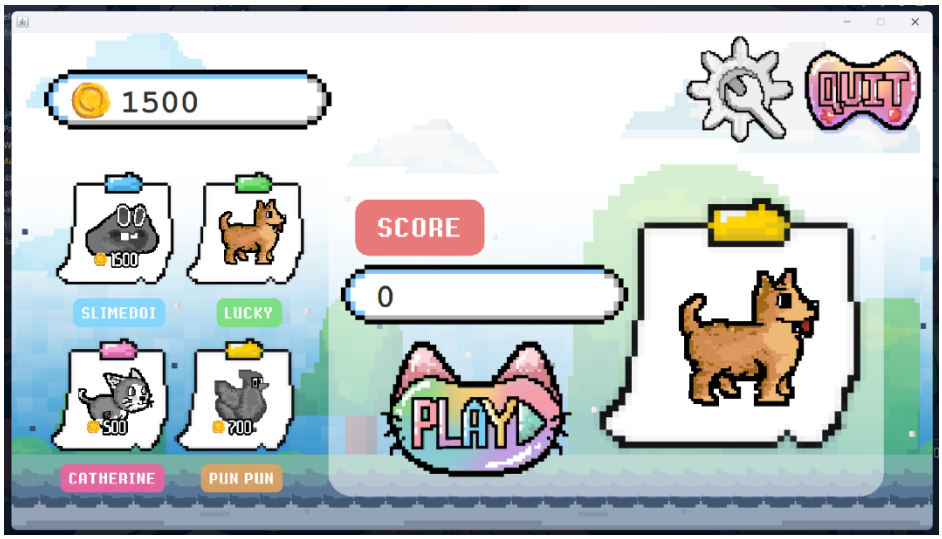
## ความสัมพันธ์ของคลาส



# บทที่ 5 ภาพการทำงานของโปรแกรม

(Screen Capture)

ภาพที่ 1 หน้าต่างรอโหลดเกม



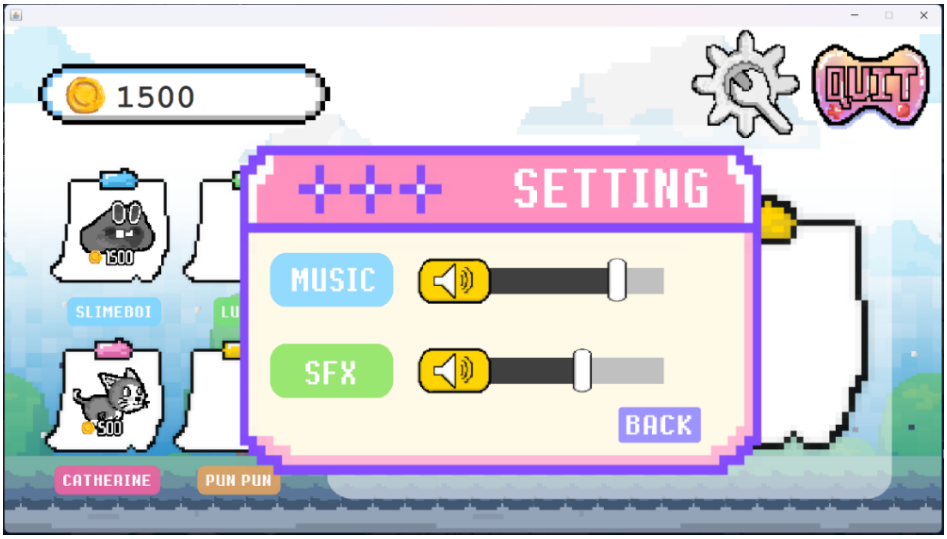

ภาพที่ 2 หน้า Lobby เลือกตัวละคร และการตั้งค่า

ภาพที่ 2.1 หน้าเลือกตัวละคร

โดยตัวละครแต่ละตัวมีราคา ดังนี้

1. LUCKY ตัวละครเริ่มต้น
2. SLIMEBOI 1500 เหรียญ
3. CATHERINE 500 เหรียญ
4. PUNPUN 700 เหรียญ

ภาพที่ 2.2 การตั้งค่า และ ออกจากเกม

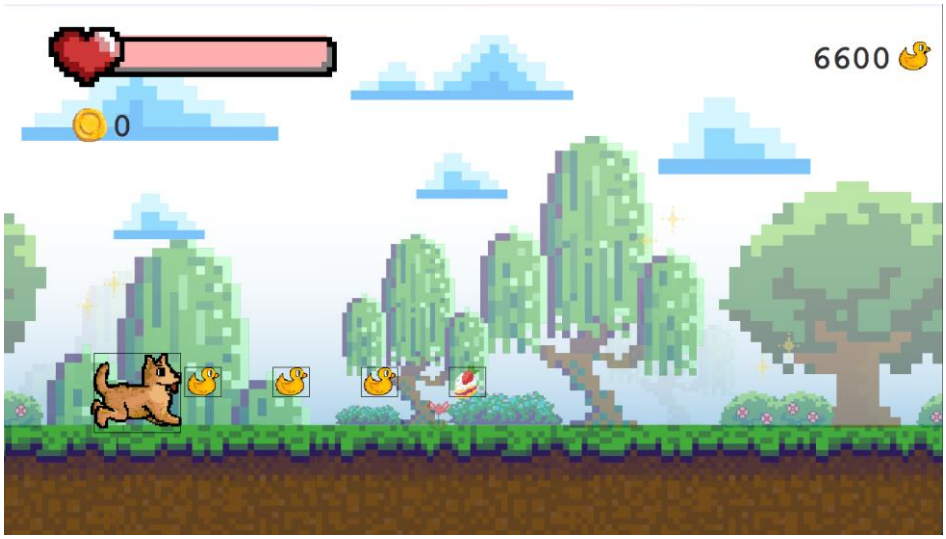



ภาพที่ 2.2.1 หน้าการตั้งค่า ควบคุมระดับเสียงเพลง และ ซาวด์เอฟเฟ็กต์ ภายในเกม

ภาพที่ 2.2.2 หน้าออกจากเกม

ภาพที่ 3  ปุ่มเริ่มเล่นเกม

ภาพที่ 4 หน้ารอเริ่มเกม




ภาพที่ 5 หน้าเกม

1. กดปุ่มลูกศรขึ้นที่แป้นพิมพ์ พื่อกระโดดข้ามสิ่งกีดขวาง
2. กดปุ่มลูกศรลงที่แป้นพิมพ์ เพื่อสไลด์หลบสิ่งกีดขวาง

ภาพที่ 6 point และ coins ที่ต้องเก็บ

ภาพที่ 7 helper Item

ภาพที่ 8 อุปสรรค